Rare Emotion Detection in Multi-Label Text Classification on GoEmotions

Literature Review

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1 Literature Review

For humans, a Reddit comment can be confusing. For a classifier asked to identify any combination of 28 possible emotion labels, it can be considerably worse. Text-based emotion detection is part of the broader field of affective computing and has been studied in natural language processing since at least the mid-2000s. In 2005, early work already explored machine-learning approaches for predicting emotions from text [2]. Two years later, the SemEval-2007 Affective Text task provided an early shared benchmark for emotion and valence prediction from news headlines [21].

By 2014, survey work had already described a range of lexical and conventional machine-learning approaches to emotion detection from text [5]. The ability to identify emotions from text is relevant to applications where systems need to interpret affective content in human language. Examples include the analysis of emotional expression in online communication, empathetic conversational systems, and the detection of harmful online behaviour [5, 10].

Over time, the field moved toward neural and transformer-based models and toward increasingly fine-grained emotion taxonomies. In 2020, GoEmotions introduced a large-scale dataset of approximately 58,000 Reddit comments annotated with 27 emotion categories and a neutral label [10]. The dataset supports fine-grained emotion recognition and includes texts associated with multiple emotions. Its original BERT baseline achieved an average F1 score of approximately 0.46, indicating that classification over the fine-grained taxonomy remains challenging.

Around the same period, research increasingly began to model dependencies between emotion labels rather than treating them as independent classes. In 2020, Gaonkar et al. [11] incorporated label semantics and learned label–label correlations into emotional reaction prediction. In 2021, SpanEmo [1] modelled associations between words and co-occurring emotion labels by formulating multi-label emotion classification as a span-prediction problem, while Seq2Emo [15] captured label dependencies through a sequence-to-emotion decoder. Together, these approaches show that relationships between emotion labels can provide useful information for multi-label emotion recognition.

Research also began to explore richer representations of the emotion label space. In 2021, Buechel et al. [4] proposed label-agnostic emotion embeddings that provide a shared latent representation across different emotion annotation schemes. In 2023, Chochlakis et al. [8, 9] further investigated label correlations and learned emotion embeddings for transferring knowledge across emotions, languages, and annotation formats. This line of work suggests that emotion labels may contain reusable structural information rather than serving only as independent output categories.

Geometric and structured approaches have made this idea more explicit. HypEmo, introduced in 2023 [6], represents hierarchical relationships between fine-grained emotions in hyperbolic space. In 2024, SEAN-GNN [25] used semantic-anchor graphs for fine-grained emotion representation learning. HyCoEM, proposed in 2025 [17], removes the need for a predefined hierarchy and instead learns semantic alignment between texts and emotion labels in hyperbolic space. On the single-label GoEmotions subset used in that work, HyCoEM reports a weighted F1 of 67.3 ± 0.5, compared with 64.0 ± 0.4 for its RoBERTa baseline. More recently, adversarial metric learning has also been applied to fine-grained emotion classification [7]. These methods primarily investigate fine-grained representation and class confusability rather than explicitly targeting rare-label performance in the original multi-label GoEmotions setting.

A second major challenge is class imbalance. In multi-label classification, rare positive labels must be learned in the presence of both frequent classes and a large number of negative label assignments. Distribution-Balanced Loss, introduced in 2020 [22], addresses long-tailed multi-label learning while accounting for distortions caused by label co-occurrence. In 2021, Asymmetric Loss [20] reduced the influence of abundant easy negative labels. In the same year, Huang et al. [16] showed that distribution-aware balancing is also effective in long-tailed multi-label text classification, where simple re-sampling can unintentionally oversample common labels that co-occur with rare ones.

Beyond loss re-weighting, long-tailed multi-label text classification has also been approached through explicit transfer of information toward rare labels. Interaction Matching [18] uses natural-language label descriptions to improve document–label matching, with much of the reported improvement coming from infrequent labels. In 2021, HTTN [23] introduced a Head-to-Tail Network that transfers meta-knowledge from data-rich head labels to data-poor tail labels. In 2023, LSFA [24] proposed label-specific feature augmentation, transferring latent feature statistics from head labels to tail labels while avoiding some of the problems caused by sample-level augmentation in multi-label data. These studies show that rare labels can benefit from information associated with better-represented labels, although they are not designed specifically around fine-grained emotion relations in GoEmotions.

The imbalance problem has also been studied directly in emotion classification. In 2025, Ramakrishnan and Dhinesh Babu [19] applied clipped asymmetric loss to BERT on GoEmotions and SemEval-2018. On GoEmotions, they report an increase in Macro-F1 from approximately 0.46 to 0.54 together with improved recognition of underrepresented emotions. This provides evidence that imbalance-aware training can improve rare-emotion performance, although the method does not learn an explicit semantic structure between emotion labels.

By 2025, several approaches had begun to combine structured representations with more difficult fine-grained or long-tailed emotion settings. TACO [13] combines structured representation learning with a long-tail objective. Its triple-view framework uses clustering-guided contrastive learning, label information, and structured instance representations to address both confusable emotions and long-tailed class distributions. On GoEmotions, TACO reports a Macro-F1 of 58.23 compared with 54.91 for its RoBERTa-base baseline and a Weighted-F1 of 65.42. Its GoEmotions experiments, however, retain only single-label examples and exclude the neutral class, so the original full multi-label task is not evaluated.

Also in 2025, Giannouris et al. [12] introduced emotion interrelation anchors for explicitly multi-label emotion detection. Their approach provides otherwise independent binary classifiers with information about relationships such as emotion co-occurrence and mutual exclusiveness. Their results show that modelling emotion interrelations can improve multi-label prediction, although rare-label recognition is not the primary objective of that work.

In 2026, GTHL-Emo [3] moved even closer to combining the two directions by addressing long-tailed multi-label emotion detection together with inter-label dependencies. Its objective combines binary cross-entropy, asymmetric focal loss, and pairwise ranking loss, while a correlation-alignment component learns label embeddings and aligns predicted label affinities with empirical co-occurrence relationships. The authors motivate this mechanism as a way of supporting rare-label prediction using information from correlated, better-represented labels. However, GTHL-Emo is evaluated on Arabic multi-label emotion datasets rather than GoEmotions.

Taken together, the literature shows that imbalance-aware learning, structured label representations, and head-to-tail information transfer are no longer separate research directions. Therefore, the remaining gap is narrower than simply combining label structure with class-imbalance handling. TACO already combines structured learning with a long-tail objective, HTTN and LSFA transfer information from better-represented labels toward tail labels, and GTHL-Emo combines imbalance-aware learning with label correlations in multi-label emotion detection.

However, these approaches differ in how label information is constructed and where it is evaluated. TACO uses a single-label subset of GoEmotions, while GTHL-Emo is evaluated on Arabic multi-label emotion datasets. General long-tailed multi-label methods such as HTTN and LSFA are not designed around the semantic structure of fine-grained emotion categories. Among the work reviewed here, a remaining question is whether complementary sources of emotion label structure can be combined and used specifically to provide stronger support for rare labels in the full 28-label multi-label GoEmotions setting.

The proposed direction is therefore a small methodological extension rather than only a comparison of existing approaches. A hybrid emotion-label representation would combine two complementary sources of information: empirical relationships observed through label co-occurrence in the training data and semantic relationships between emotion labels. The use of this structure would then be made tail-aware, so that labels with fewer training examples receive stronger structural support than well-represented labels. This makes it possible to test whether adapting structured label information to label frequency provides an additional benefit beyond conventional imbalance-aware training and non-tail-aware structured learning.

The novelty of the proposed approach lies in combining empirical and semantic emotion-label structure with frequency-dependent weighting specifically for rare-label support in the full multi-label GoEmotions setting.

This leads to the following research question:

> *Can a tail-aware combination of empirical and semantic emotion-label relationships improve rare-emotion recognition in the full 28-label multi-label GoEmotions setting beyond imbalance-aware training and non-tail-aware structured learning, without substantially degrading overall classification performance?*

Evaluation should consider both overall performance and behaviour on rare labels. Hinojosa Lee et al. [14] showed that the choice between Micro-F1, Macro-F1, and Weighted-F1 affects how multi-label emotion classification performance is interpreted under imbalance. Global F1 measures should therefore be complemented with tail-specific precision, recall, and F1 when evaluating whether improvements on rare emotions come at the cost of overall classification quality.

Use of Generative AI

Generative AI tools were used as support throughout the literature review process. They assisted with searching for potentially relevant publications, organizing and comparing papers, identifying closely related work, examining the proposed research gap against the reviewed literature, checking bibliographic and factual details, and improving the structure and wording of the text.

The author defined the overall research topic and experimental direction, reviewed and made the final decisions about paper selection, evaluated the suggested comparisons and research gap, and made the final decisions about the research question and proposed methodological direction. The author also reviewed and approved the final text. Generative AI output was not treated as a primary source; factual claims and bibliographic information included in the review were checked against the corresponding publications.

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